Type 2 Diabetes: Causes, Diagnosis, Management and Prevention

An Educational Review for Patients, Students, and Healthcare Workers

Sample document prepared for the ResearchMate RAG project

Abstract

Type 2 diabetes mellitus is a chronic metabolic disease in which the body either does not produce enough insulin or cannot use insulin effectively. This leads to persistently high blood glucose levels. Over many years, uncontrolled glucose damages blood vessels, nerves, the eyes, the kidneys, and the heart.

This review summarizes the causes, risk factors, symptoms, diagnosis, complications, treatment, and prevention of type 2 diabetes. It is written in a simple, structured format so that patients, students, and healthcare workers can understand the main ideas and find answers to common questions.

The central message is that type 2 diabetes is largely manageable and often preventable. Early detection, healthy eating, regular physical activity, weight control, medication when needed, and regular follow-up can greatly reduce the risk of serious complications and improve quality of life.

1. Introduction

Diabetes mellitus is a group of metabolic disorders characterized by high blood glucose. Glucose is the main fuel for the body's cells, and it comes mostly from the carbohydrates we eat. To enter cells and be used for energy, glucose needs the help of a hormone called insulin.

Insulin is produced by the beta cells of the pancreas. After a meal, blood glucose rises, and the pancreas releases insulin. Insulin signals muscle, fat, and liver cells to absorb glucose from the blood, which brings the blood glucose level back to its normal range.

There are several types of diabetes. Type 1 diabetes is an autoimmune condition in which the immune system destroys insulin-producing cells. Gestational diabetes appears during pregnancy. Type 2 diabetes, the most common form, accounts for the large majority of all diabetes cases worldwide.

Type 2 diabetes usually develops slowly and often in adults, but it is increasingly diagnosed in adolescents and young adults because of rising obesity and sedentary lifestyles. Because early symptoms are mild, many people live with the disease for years before it is detected.

The disease matters not only because of high glucose itself but because of the long-term damage it causes. Diabetes is a leading cause of blindness, kidney failure, lower-limb amputation, heart attack, and stroke, which makes prevention and early management a major public health priority.

This document is intended for educational purposes. It does not replace advice from a qualified doctor. Anyone who suspects they have diabetes, or who is already diagnosed, should consult a healthcare professional for personal diagnosis, treatment, and follow-up.

2. Epidemiology and Burden

Diabetes has become one of the fastest-growing health problems in the world. According to estimates from the International Diabetes Federation, more than half a billion adults were living with diabetes in 2021, and the number is projected to keep increasing over the coming decades.

A large share of people with diabetes are undiagnosed. Because type 2 diabetes can be silent for years, many individuals discover it only when a complication appears, such as vision problems, a foot ulcer that does not heal, or chest pain.

South Asia carries a particularly heavy burden. Genetic susceptibility, central obesity at relatively low body weight, changing diets, urbanization, and reduced physical activity have all contributed to high rates of diabetes in countries such as Pakistan, India, and Bangladesh.

Pakistan is among the countries with the highest diabetes prevalence in the world. IDF estimates from 2021 suggested that roughly one in four adults in Pakistan was living with diabetes, which equals tens of millions of people and places great pressure on the health system.

The economic burden of diabetes is significant. Costs include medicines, test strips, hospital stays, treatment of complications, and lost productivity. In low-income settings, families often pay for care out of pocket, and diabetes can push households deeper into poverty.

Diabetes also increases the risk of death from other causes. People with diabetes have a higher chance of dying from cardiovascular disease and kidney disease, and they may be more vulnerable to serious infections, which is why overall health monitoring is important.

3. Pathophysiology

The main problem in type 2 diabetes is a combination of insulin resistance and relative insulin deficiency. Insulin resistance means that muscle, fat, and liver cells respond poorly to insulin. The pancreas must therefore produce more insulin to keep blood glucose normal.

In the early stage, the pancreas compensates by releasing larger amounts of insulin. Blood glucose may remain normal or only slightly high during this period, even though the underlying problem is already present. This stage is sometimes called prediabetes when glucose begins to rise.

Over time, the beta cells become exhausted and lose their ability to keep up with the demand. Insulin production falls, and blood glucose rises above the normal range. At this point the person meets the criteria for diabetes.

The liver plays an important role. In healthy people, insulin tells the liver to stop releasing glucose after meals. In insulin resistance, the liver continues to release glucose into the blood, which explains why fasting blood glucose is often high in type 2 diabetes.

Excess fat, especially visceral fat stored around the abdomen, is closely linked to insulin resistance. Fat tissue releases inflammatory substances and free fatty acids that interfere with insulin signaling, making it harder for cells to absorb glucose.

Muscle tissue is the largest site of glucose uptake in the body. Physical inactivity reduces the number and activity of glucose transporters in muscle, while regular exercise improves insulin

sensitivity, which is one reason activity is a cornerstone of treatment.

Hormones from the gut, called incretins, help regulate insulin release after meals. In type 2 diabetes, the incretin effect is reduced. Several modern medicines work by enhancing the action of incretin hormones to improve insulin secretion and lower glucose.

Chronic high glucose is itself harmful to beta cells and insulin-sensitive tissues, a concept known as glucotoxicity. Elevated blood fats can also damage cells, known as lipotoxicity. These effects create a vicious cycle that worsens the disease if it is not treated.

Genetics influence susceptibility. Having a parent or sibling with type 2 diabetes raises a person's risk considerably. However, genes alone do not explain the disease, and environmental factors such as diet and activity determine whether the genetic risk becomes actual disease.

Understanding these mechanisms explains the logic of treatment. Weight loss and exercise reduce insulin resistance, some medicines improve insulin sensitivity, others increase insulin release, and others reduce glucose absorption or promote glucose loss in the urine.

4. Risk Factors

Risk factors for type 2 diabetes are divided into those that cannot be changed and those that can be modified. Non-modifiable factors include age, family history, and ethnic background. Modifiable factors include body weight, diet, physical activity, and smoking.

Age is an important risk factor. The risk rises steadily after the age of 40 or 45, although the disease is now seen in younger people as well. Regular screening is therefore recommended for adults beyond middle age even when they feel completely well.

A family history of diabetes in a parent or sibling substantially increases the chance of developing the disease. People with such a history should be especially careful about weight and lifestyle and should be tested more regularly.

Overweight and obesity are the strongest modifiable risk factors. The risk is highest when fat is concentrated around the waist. In South Asian populations, diabetes can develop at lower body mass index values than in other populations, so waist size is a useful warning sign.

A sedentary lifestyle increases risk independently of body weight. Long hours of sitting, limited walking, and lack of structured exercise all contribute. Even moderate activity such as brisk walking for about thirty minutes on most days can lower the risk.

Unhealthy diet patterns also raise risk. Frequent consumption of sugary drinks, refined carbohydrates such as white bread and white rice in large portions, fried foods, processed meat, and sweets contributes to weight gain and poor glucose control.

Prediabetes is a major warning stage. People with prediabetes have blood glucose levels higher than normal but not yet high enough to be called diabetes. Without lifestyle change, a large proportion of them progress to type 2 diabetes within several years.

Women who had gestational diabetes during pregnancy, or who delivered a very large baby, have a higher lifetime risk of type 2 diabetes. Polycystic ovary syndrome is also associated with insulin resistance and an increased risk of diabetes.

High blood pressure and abnormal blood fats, such as high triglycerides and low HDL cholesterol, often occur together with insulin resistance. This cluster of problems is called the metabolic syndrome and strongly predicts diabetes and heart disease.

Other contributors include smoking, poor sleep, chronic stress, and the long-term use of certain medicines such as corticosteroids. Addressing these factors, along with diet and exercise, forms part of a complete approach to reducing risk.

5. Symptoms and Clinical Presentation

Many people with early type 2 diabetes have no symptoms at all. The disease is frequently found during a routine blood test or while evaluating another problem. This silent nature is the main reason regular screening is so valuable.

When blood glucose becomes high enough, classic symptoms appear. These include frequent urination, excessive thirst, and increased hunger. High glucose spills into the urine and draws water with it, which causes dehydration and the need to drink more fluids.

Unexplained weight loss can occur when the body cannot use glucose properly and begins to break down fat and muscle for energy. In type 2 diabetes, weight loss is less dramatic than in type 1, but it may still be present in poorly controlled disease.

Fatigue and weakness are common because cells are not receiving enough glucose for energy. People may also notice difficulty concentrating, irritability, or a general feeling of being unwell that does not improve with rest.

Blurred vision can occur when high glucose changes the shape of the lens in the eye. This symptom may come and go with glucose levels. Persistent vision changes should always be checked by an eye doctor because they may indicate damage to the retina.

Slow healing of cuts and wounds, recurring skin infections, and frequent fungal infections such as genital thrush are common clues. High glucose weakens the immune response and reduces blood flow, so infections develop more easily and heal more slowly.

Tingling, numbness, burning pain, or loss of sensation in the feet and hands may indicate diabetic neuropathy. This symptom often appears after years of high glucose, but it can be present at the time of diagnosis in people whose diabetes went undetected for long.

A dark, velvety thickening of the skin on the neck, armpits, or groin, called acanthosis nigricans, is a visible sign of insulin resistance. It is commonly seen in people with obesity and is a useful clue for doctors to look for diabetes or prediabetes.

6. Diagnosis

Diabetes is diagnosed through blood tests that measure glucose or glycated hemoglobin. Diagnosis should not be made from a single home glucometer reading. Laboratory-confirmed tests are used so that treatment decisions are based on reliable results.

The fasting plasma glucose test measures blood glucose after at least eight hours without food. A fasting value of 126 mg/dL (7.0 mmol/L) or higher indicates diabetes. A value from 100 to 125 mg/dL is classified as impaired fasting glucose, a form of prediabetes.

The oral glucose tolerance test measures blood glucose two hours after drinking a standard glucose solution. A two-hour value of 200 mg/dL (11.1 mmol/L) or higher indicates diabetes. A value from 140 to 199 mg/dL indicates impaired glucose tolerance.

The HbA1c test reflects average blood glucose over the previous two to three months. An HbA1c of 6.5 percent or higher indicates diabetes, while 5.7 to 6.4 percent indicates prediabetes. This test does not require fasting, which makes it convenient for patients.

A random plasma glucose of 200 mg/dL or higher, in a person with classic symptoms such as excessive thirst and frequent urination, is also diagnostic. In the absence of symptoms, an abnormal test is usually repeated on another day to confirm the diagnosis.

Some conditions can make HbA1c unreliable, including anemia, recent blood loss or transfusion, and certain hemoglobin variants. In such cases, doctors rely on glucose-based tests. The choice of test should be guided by the individual clinical situation.

Screening is recommended for adults at higher risk. This includes people who are overweight with additional risk factors, those with a family history, women with previous gestational diabetes, and everyone beyond a certain age, commonly 35 to 45 years, even without symptoms.

At diagnosis, doctors usually perform additional tests to assess overall health. These may include blood pressure measurement, a lipid profile, kidney function tests, a urine test for protein, an eye examination, and a foot examination, because complications may already be present.

Self-monitoring of blood glucose with a glucometer helps patients understand how food, activity, stress, and medicines affect their glucose. Some patients use continuous glucose monitoring devices, which provide readings throughout the day and night.

Correct diagnosis requires distinguishing type 2 diabetes from type 1 and other forms. Young age, rapid weight loss, ketones in the urine, and very high glucose at presentation may suggest type 1 diabetes, which requires insulin treatment from the start.

7. Complications

Complications of diabetes are divided into microvascular complications, which affect small blood vessels, and macrovascular complications, which affect large blood vessels. Both result from long exposure to high glucose together with other risk factors such as high blood pressure.

Diabetic retinopathy is damage to the small blood vessels of the retina at the back of the eye. In its early stages it causes no symptoms. Without treatment it can progress to severe vision loss, and it is one of the leading causes of blindness in working-age adults.

Regular dilated eye examinations allow retinopathy to be detected early. People with type 2 diabetes should have an eye examination at diagnosis and then at regular intervals. Treatments such as laser therapy and injections can protect vision when started in time.

Diabetic nephropathy is kidney damage caused by diabetes. The kidney filters blood through tiny vessels that are vulnerable to high glucose and high blood pressure. Early damage appears as small amounts of protein, called albumin, in the urine.

If nephropathy progresses, kidney function declines and may lead to chronic kidney disease and eventually kidney failure requiring dialysis or transplantation. Annual urine albumin and blood

creatinine tests help detect kidney problems at a stage when treatment is most effective.

Diabetic neuropathy is nerve damage that most commonly affects the feet and legs. It causes numbness, tingling, or burning pain. Loss of sensation is dangerous because injuries may go unnoticed, allowing small wounds to become serious infections.

Autonomic neuropathy affects nerves that control internal organs. It can cause problems such as delayed stomach emptying, dizziness on standing, abnormal heart rate, bladder problems, and sexual dysfunction. These symptoms are often under-recognized and can reduce quality of life.

Diabetic foot disease results from a combination of neuropathy, poor circulation, and infection. Foot ulcers can develop from minor trauma such as ill-fitting shoes. Severe cases may require amputation, which is why daily foot inspection and proper footwear are essential.

Cardiovascular disease is the leading cause of death among people with diabetes. High glucose, high blood pressure, and abnormal lipids accelerate atherosclerosis, increasing the risk of heart attack, stroke, and peripheral arterial disease.

Heart disease in diabetes can sometimes be silent. Due to nerve damage, a person may not feel typical chest pain during a heart attack. Doctors therefore pay close attention to blood pressure, cholesterol, and other cardiovascular risk factors in all patients with diabetes.

Acute complications can be life-threatening. Severe hyperglycemia may cause a hyperosmolar hyperglycemic state with extreme dehydration, particularly in older adults. Hypoglycemia, or low blood glucose, can occur as a side effect of certain diabetes medicines, especially insulin and sulfonylureas.

Symptoms of hypoglycemia include sweating, shaking, rapid heartbeat, hunger, confusion, and in severe cases loss of consciousness. Patients on glucose-lowering drugs should know how to recognize hypoglycemia and treat it promptly with a fast-acting source of sugar.

8. Lifestyle and Nutrition Management

Lifestyle change is the foundation of type 2 diabetes treatment, and it remains important even when medicines are needed. Improvements in diet, physical activity, and body weight can lower blood glucose, reduce the need for medicines, and in some people lead to remission.

Medical nutrition therapy aims to achieve healthy body weight, stable blood glucose, and good control of blood pressure and cholesterol. There is no single diet for everyone. The best plan is one that is nutritious, culturally acceptable, affordable, and sustainable for the individual.

Carbohydrate quality and quantity both matter. Whole grains, legumes, vegetables, and fruits provide fiber that slows glucose absorption. Refined grains, sugary drinks, sweets, and large portions of white rice or bread cause faster glucose rises and should be limited.

The glycemic index ranks foods by how quickly they raise blood glucose. Foods with a lower glycemic index, such as lentils, chickpeas, oats, and most non-starchy vegetables, tend to produce a gentler rise in glucose than foods such as white bread or sweetened cereals.

Portion control is a practical tool. Many people find the plate method helpful, in which half of the plate is filled with non-starchy vegetables, one quarter with protein such as fish, chicken, eggs, or pulses, and one quarter with carbohydrate such as roti or rice.

Healthy fats should replace unhealthy fats. Sources such as nuts, seeds, olive oil, and fish are preferable to ghee in large amounts, deep-fried foods, and processed snacks. Reducing salt intake also helps control blood pressure, which is important in people with diabetes.

Regular physical activity increases insulin sensitivity and helps control weight. Adults are generally advised to aim for at least 150 minutes of moderate aerobic activity per week, such as brisk walking, cycling, or swimming, spread over several days.

Resistance exercise, such as lifting weights or using body weight, builds muscle and further improves glucose control. Guidelines suggest including resistance training two or three times a week, along with reducing long periods of uninterrupted sitting.

Weight loss has powerful effects. Losing even five to ten percent of body weight can improve glucose, blood pressure, and cholesterol. In some people with recent-onset type 2 diabetes, substantial weight loss can lead to remission, meaning glucose returns to near-normal levels.

Other healthy habits include quitting smoking, limiting alcohol, getting enough sleep, and managing stress. Education and psychological support are also important, since living with a chronic disease can cause worry, fatigue, and a feeling of burnout known as diabetes distress.

9. Medications and Monitoring

When lifestyle measures alone are not enough to reach glucose targets, medicines are added. The choice of medicine is individualized and depends on factors such as glucose level, presence of heart or kidney disease, risk of hypoglycemia, weight, side effects, and cost.

Metformin is usually the first medicine prescribed for type 2 diabetes unless there is a reason to avoid it. It reduces glucose production by the liver and improves insulin sensitivity. It rarely causes hypoglycemia, and common side effects are stomach upset and diarrhea.

Sulfonylureas stimulate the pancreas to release more insulin. They are effective and inexpensive, but they can cause hypoglycemia and weight gain. Patients taking them should eat regularly and should be aware of the symptoms of low blood glucose.

DPP-4 inhibitors and GLP-1 receptor agonists act on the incretin system. GLP-1 receptor agonists can also reduce appetite and body weight, and some have been shown to lower the risk of cardiovascular events in people at high risk.

SGLT2 inhibitors lower glucose by causing the kidneys to remove glucose through the urine. They may also reduce body weight and blood pressure, and certain medicines in this class protect the heart and kidneys. Possible side effects include genital infections and dehydration.

Insulin is used when other medicines are not sufficient or when glucose is very high. Many people with long-standing type 2 diabetes eventually need insulin because of declining beta cell function. Insulin therapy is safe and effective when used with proper education and monitoring.

Treatment goals are individualized. A common target for many adults is an HbA1c of around 7 percent, but a doctor may set a higher or lower goal depending on age, other illnesses, and the risk of hypoglycemia. Blood pressure and cholesterol targets are equally important.

Regular follow-up is essential. HbA1c is commonly checked every three to six months, while kidney function, urine albumin, cholesterol, eyes, and feet are checked at least yearly. Vaccinations, such

as influenza and pneumococcal vaccines, are also recommended for many patients.

10. Prevention and Public Health

Type 2 diabetes is largely preventable, especially in people with prediabetes. Large clinical trials have shown that structured lifestyle programs focusing on modest weight loss and regular physical activity can reduce the risk of developing diabetes by more than half.

Prevention begins with identifying people at risk. Simple risk scores based on age, waist circumference, family history, and physical activity can be used in community settings to decide who should have a blood glucose test, even in places with limited laboratory facilities.

Healthy eating habits should start early in life. Families can reduce sugary drinks, offer fruits and vegetables, avoid excessive fried and processed food, and encourage children to be physically active. Healthy habits learned in childhood often continue into adulthood.

Schools and workplaces can support prevention. Providing healthy food options, safe spaces for physical activity, and regular health check-ups helps people maintain a healthy weight and detect problems early, before complications develop.

Public health policy can have a strong influence. Measures such as taxes on sugary beverages, clear food labeling, restrictions on marketing unhealthy foods to children, and urban planning that encourages walking and cycling may reduce diabetes risk across the whole population.

Primary healthcare services play a central role in prevention and early care. Trained community health workers can screen for risk factors, provide lifestyle counseling, and refer patients to doctors, which extends diabetes care to people who live far from hospitals.

Health education and awareness campaigns are valuable, particularly where many people are unaware of their disease. Messages should be simple and practical, delivered in local languages, and should explain that diabetes can be controlled and that complications can be avoided.

Pregnancy offers a special opportunity. Women with gestational diabetes should receive follow-up testing after delivery, because they have a high chance of developing type 2 diabetes later. Healthy eating, breastfeeding, and regular activity can lower this risk.

11. Challenges in Low-Resource Settings

In many low- and middle-income countries, including Pakistan, access to quality diabetes care is uneven. People in rural areas may live far from clinics, and specialist doctors, diagnostic laboratories, and eye and foot screening services are often concentrated in large cities.

The cost of medicines, glucose test strips, and regular laboratory tests can be a heavy burden, especially when patients pay out of pocket. Some people stop treatment when money runs short, which leads to poor control and avoidable complications.

Low health literacy and misinformation are important challenges. Some patients believe that diabetes can be cured by herbal remedies alone or that medicines should be stopped once glucose improves. Clear communication from healthcare providers is needed to correct such beliefs.

Cultural food habits, social gatherings, and limited safe places for exercise, especially for women, can make lifestyle change difficult. Care plans should be realistic and respectful of local culture, and should involve the family in supporting the patient.

Digital tools can help overcome some barriers. Reliable information sources, reminder systems, and question-answering assistants can support patient education and help healthcare workers find guidelines quickly, as long as the information is accurate and clearly based on trusted documents.

12. Conclusion

Type 2 diabetes is a common, serious, but manageable condition. It develops through a combination of insulin resistance and declining insulin production, influenced by genetics, body weight, diet, and physical activity. Early symptoms are often absent, so screening is essential.

Diagnosis relies on laboratory tests such as fasting glucose, oral glucose tolerance, and HbA1c. Once diagnosed, the aim of care is to keep glucose, blood pressure, and cholesterol in a safe range to prevent damage to the eyes, kidneys, nerves, and cardiovascular system.

Healthy eating, regular exercise, weight management, and appropriate medicines form the core of treatment, supported by regular monitoring and patient education. Prevention programs targeting people with prediabetes can delay or avoid the onset of the disease.

With timely detection, good self-care, and consistent medical follow-up, people with type 2 diabetes can live long and healthy lives. Strengthening awareness, screening, and affordable care remains one of the most important health priorities for communities and health systems.